

When EOS Tokens Disagree: Understanding Length Inflation in On-Policy Distillation

Yuxiao Yang^{1,*}, Tianrun Yu^{2,*}, Shangzhe Li^{1,*}, Kaixiang Zhao^{2,*}, Xuchao Zhang³, Chetan Bansal³, Huaxiu Yao¹, Taylor W. Killian^{2,†}, Weitong Zhang^{1,‡}

¹University of North Carolina at Chapel Hill, ²Brigham Young University, ³Microsoft

*Equal contribution., [†]tkillian@cs.byu.edu, [‡]weitongz@unc.edu

We study length inflation in on-policy distillation (OPD), where student responses can become excessively long and even exhaust the generation budget. We identify *termination-token mismatch* between base students and post-trained teachers as an important source of this behavior. Across Qwen3, Llama, and Gemma, the two models can place their stopping probability on different EOS tokens, even when their declared stopping sets are identical. This mismatch can suppress the student’s preferred termination action without reliably transferring the teacher-preferred alternative. We show that aligning the decoding stopping set alone is insufficient, while treating functionally equivalent EOS tokens as a shared semantic stopping action substantially mitigates mismatch-induced length inflation across all three model families. To further understand how termination behavior evolves over training, we study OPD across different K2-Horizon training stages. This stage-wise analysis shows that termination preferences can shift substantially during training, while also revealing a distinct length inflation late in the OPD run that persists beyond termination alignment. Together, these results identify termination mismatch as an important, but not exhaustive, source of OPD length dynamics. We release an implementation incorporating the proposed termination-handling corrections.

Project Page: <https://uncsciml.github.io/opd-eos-website>

1 Introduction

On-policy distillation (OPD) trains a student using its own rollouts with dense token-level supervision from a teacher model (Lu and Lab, 2025). It has recently attracted growing attention as a practical framework for transferring strong post-trained behavior to smaller or less capable models (Jin et al., 2026; Yang et al., 2026a; Li et al., 2026). Despite its effectiveness, several recent studies have reported undesired length inflation in student responses, often accompanied by truncation, repetition, or other generation pathologies (Luo et al., 2026). We observe the same phenomenon across several model families, using pretrained base students and post-trained teachers in single-turn mathematical reasoning. Student responses become progressively longer during training, and in severe cases a large fraction of rollouts exhaust the generation budget. We refer to this behavior as *length inflation*. Figure 1 illustrates a Qwen3 example in which the student reaches the correct answer early but continues generating repetitive or redundant suffixes instead of terminating. Such increases in response length need not reflect more extensive reasoning; they can instead arise from a failure to stop. Existing studies have proposed different explanations and mitigation strategies for this phenomenon, ranging from objective-level effects of reverse-KL distillation to rollout degradation and other training instabilities (Fu et al., 2026; He et al., 2026; Luo et al., 2026; Wang et al., 2026; Zhao et al., 2026).

In this work, however, we find that a direct cause of this inflation in the configurations studied is *termination-token mismatch* between base students and post-trained teachers. We observe this mismatch across Qwen3, Llama, and Gemma. The key issue is that the student and teacher may represent the same semantic decision to terminate using different EOS tokens. This mismatch can arise not only when their declared decoding stopping sets differ, but also when the two models assign their learned stopping probability to different tokens within the same declared set. Since the teacher only evaluates the tokens generated by students in OPD, a student-generated EOS token receives a negative distillation signal if teacher model is actively using an alternative EOS token. The student’s existing termination action can therefore be repeatedly suppressed without reliably transferring the teacher’s termination token with minor ($\sim 10^{-11}$ in Qwen3) in student model.

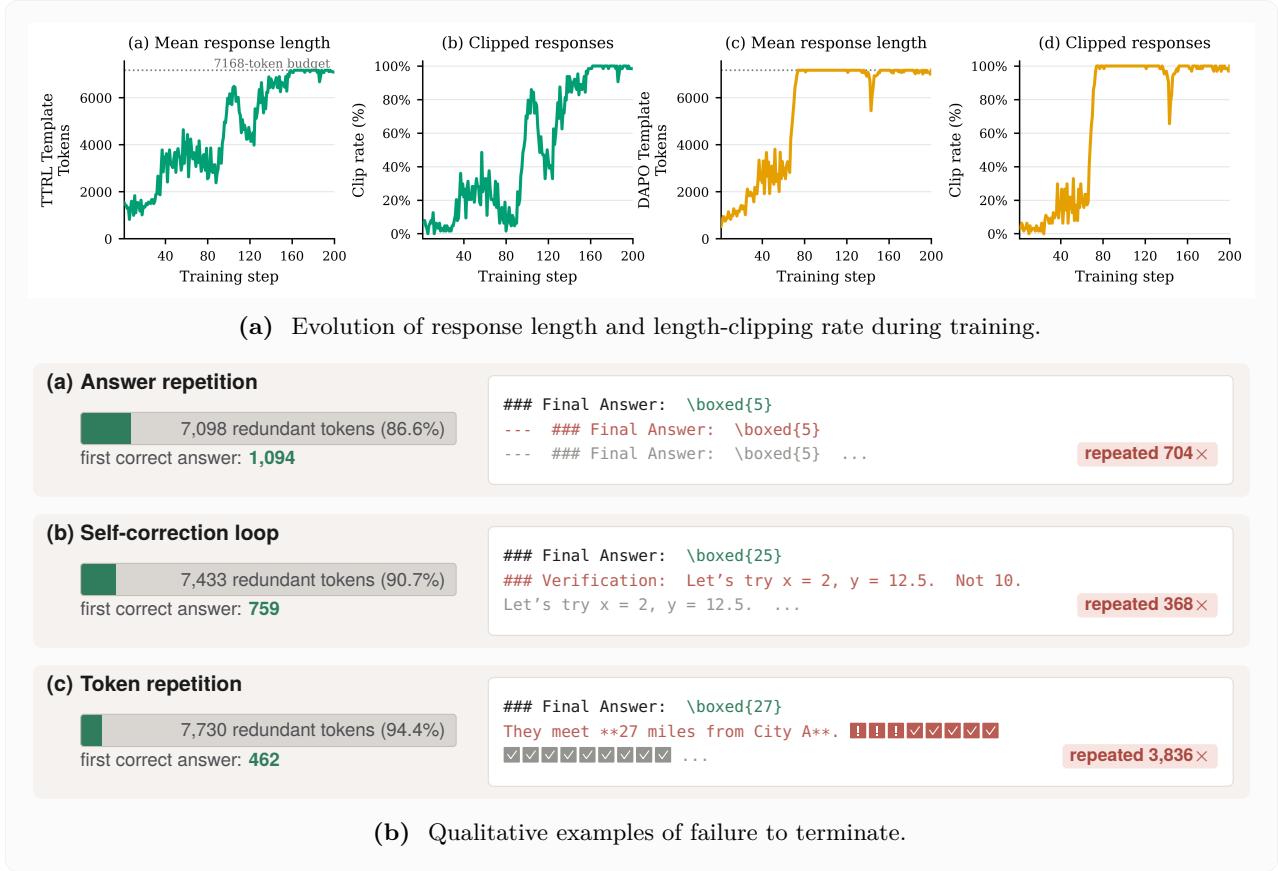


Figure 1 Length inflation under vanilla OPD, illustrated with Qwen3. (a) When distilling a post-trained Qwen3 teacher into a Qwen3 base student, response length increase and hits the generation budget during training under both TTRL and DAPO prompts. (b) Inflated responses often contain the correct answer early but fail to terminate, consuming much of the remaining generation budget with repetitive or redundant continuations.

To address this mismatch, we first show that aligning the decoding interface alone is insufficient. We then compare three corrections that align termination in the distillation signal and find that treating functionally equivalent EOS tokens as a shared semantic stopping action substantially mitigates mismatch-induced length inflation across Qwen3, Llama, and Gemma in single-turn mathematical reasoning.

To further understand how this termination mismatch evolves over training, we leverage the intermediate K2-Horizon checkpoints spanning pretraining, midtraining, supervised fine-tuning, and final post-training. This stage-wise analysis shows how learned termination preferences shift across training stages, but also reveals a distinct inflation late in the OPD run: even after the teacher-preferred termination form has been transferred, response length can increase again and termination probability can collapse. This behavior is not explained by the EOS mismatch alone and points to additional mechanisms governing OPD length dynamics. We also observe that prompt-template choices can further modulate response length and measured performance.

In short, beyond the mechanisms considered in prior work, we identify termination-token mismatch between base and post-trained models as a major and readily diagnosable source of length inflation. Our contributions are summarized as follows:

- We identify termination-token mismatch as an important mechanism of length inflation in sampled-token OPD. Across Qwen3, Llama, and Gemma, we show that student and teacher checkpoints can represent the same semantic stopping decision through different learned token preferences, even when their declared EOS sets are identical.
- We systematically compare four termination-handling strategies on Qwen3 and show that aligning the

decoding stopping set alone is insufficient. Corrections that align termination semantics in the distillation signal substantially mitigate mismatch-induced inflation, and we validate semantic EOS aggregation across Qwen3, Llama, and Gemma.

- We extend the analysis across post-training stages using K2-Horizon, showing how learned termination preferences evolve during training and revealing a distinct length inflation late in the OPD run from the pretraining checkpoint (Phase 3 in Figure 7), which remains after termination alignment. We also document sensitivity to prompt and evaluation protocols, highlighting these choices as an additional source of variation in reported OPD behavior.
- We release an implementation that includes the termination-handling corrections and the evaluation protocols.

2 Experimental Setup

Sampled-token OPD and notation. On-policy distillation trains a student π_θ on its own rollouts with a fixed teacher π^E . At a prefix $s_t = (x, y_{<t})$, the sampled token y_t receives the signal

$$A_t = \log \pi^E(y_t | s_t) - \log \pi_\theta(y_t | s_t), \quad g_t = A_t \nabla_\theta \log \pi_\theta(y_t | s_t).$$

We treat A_t as detached when we compute the score-function update, and we average the contributions of all valid generated tokens without reward-to-go. We refer to this formulation as sampled-token OPD, in which the teacher scores only the token that the student has sampled. We write $\mathcal{E}_{\text{EOS}}$ for the set of termination tokens that are equivalent under the rollout protocol of a given model pair, and $e_\star \in \mathcal{E}_{\text{EOS}}$ for the termination token that the base student supports natively. The corrections in Section 3 all act on these two objects. Our experiments build on the public OPD implementation of Li et al. (2026).

Models and data. Our main pair uses Qwen3-1.7B-Base as the student and Qwen3-4B in non-thinking mode as the teacher, and we run the full ablation over the four termination-handling strategies on this pair. To test whether the mechanism extends beyond Qwen3, we additionally use Llama-3.2-3B Base and Instruct (Grattafiori et al., 2024) and Gemma-3-4B PT and IT (Team et al., 2025). Within each pair, the student and the teacher share a tokenizer and a vocabulary. To observe how termination preferences change during post-training, we also use K2-Horizon-7B, which releases checkpoints from pretraining, midtraining, supervised fine-tuning, and the final post-trained model. We fix the final checkpoint as the teacher and initialize the student from each of the earlier stages. All models are trained on DAPO-Math-17K (Yu et al., 2026). Unless otherwise specified, both training and evaluation use the TTRL template, which requires the final answer to be enclosed in `\boxed{\}`. The template comparisons additionally use the DAPO template and a raw-question template that provides only the problem statement. Prompt strings and grader details are provided in Section B.

Metrics and budgets. During training we track the mean response length and the clipping ratio, which is the fraction of responses that reach the generation-length budget. For downstream performance we report Avg@16, which samples 16 responses per problem, averages correctness over the problems in each benchmark, and then takes the unweighted mean over AMC23 (Mathematical Association of America, 2023), AIME24 (Zhang and Math-AI, 2024), and AIME25 (Zhang and Math-AI, 2025). To observe termination behavior directly, we also compare the termination probabilities of the student and the teacher at the final position of each student rollout, where we report both the probabilities of individual tokens and the total stopping mass $q_\pi(h) = \sum_{e \in \mathcal{E}_{\text{EOS}}} \pi(e | h)$. These prefixes include rollouts that end naturally and rollouts that are truncated at the budget. The total stopping mass measures how strongly a model prefers to stop at a given prefix, which is a different quantity from the fraction of rollouts that terminate naturally. In the main experiments, training allows at most 1,024 prompt tokens and 7,168 response tokens, while independent checkpoint evaluation allows at most 8,192 generated tokens, and the settings for each K2-Horizon stage are described with the corresponding experiments. The teacher reference lines in our figures report the response length and clipping statistics of teacher generations on the same prompts, which is a different quantity from the teacher probabilities that we evaluate at student prefixes. Remaining optimization and sampling settings are provided in Section A.

Table 1 EOS tokens of model studied. We report the `eos_token_id` specified in each model’s `generation_config.json`. Models sharing the same declared EOS token ID set are shown in the same color; token IDs are given in parentheses. For IFM/K2-Horizon-7B, revision names identify the training-stage models used in our experiments. Its EOS token strings were renamed after pretraining, while their IDs and the declared stopping set remained unchanged.

Family	Stage	Version	Checkpoint(s)	EOS token(s)
Qwen	Base	Qwen2.5-Math	Qwen2.5-Math-1.5B Qwen2.5-Math-7B	< endoftext > (151643)
		Qwen3	Qwen3-1.7B-Base Qwen3-4B-Base	
	Post-trained	Qwen2.5-Math	Qwen2.5-Math-1.5B-Instruct Qwen2.5-Math-7B-Instruct	< im_end >, < endoftext > (151645, 151643)
		Qwen3	Qwen3-1.7B Qwen3-4B	
Llama	Base	Llama 3.1	Llama-3.1-8B Llama-3.1-70B	< end_of_text > (128001)
		Llama 3.2	Llama-3.2-1B Llama-3.2-3B	
	Post-trained	Llama 3.1	Llama-3.1-8B-Instruct Llama-3.1-70B-Instruct	< end_of_text >, < eom_id >, < eot_id > (128001, 128008, 128009)
		Llama 3.2	Llama-3.2-1B-Instruct Llama-3.2-3B-Instruct	
Gemma	Base	Gemma 2	gemma-2-2b	<eos> (1) <eos>, <end_of_turn> (1, 106)
		Gemma 3	gemma-3-1b-pt gemma-3-4b-pt	
	Post-trained	Gemma 2	gemma-2-2b-it	<eos>, <end_of_turn> (1, 107) <eos>, <end_of_turn> (1, 106)
		Gemma 3	gemma-3-1b-it gemma-3-4b-it	
K2-Horizon	Base	K2-Horizon	K2-Horizon-7B pretrain_final	< endoftext >, < im_end > (1, 250019)
	Mid-training	K2-Horizon	K2-Horizon-7B mid_1_final	< ifm endoftext >, < ifm im_end > (1, 250019)
	Post-trained	K2-Horizon	K2-Horizon-7B sft_1_final main	< ifm endoftext >, < ifm im_end > (1, 250019)

3 Termination Mismatch and Probability-Level Alignment

We use Qwen3 to diagnose termination-token mismatch. We first show that aligning the decoding stopping set does not remove length inflation, then analyze how the mismatch enters sampled-token supervision, then compare three corrections that align termination at the probability level, and finally test the mechanism on Llama 3.2 and Gemma 3.

3.1 Decoding EOS Alignment Alone Is Insufficient

As shown in Figure 1a, vanilla OPD on Qwen3 progressively increases response length and clipping, and many rollouts eventually exhaust the generation budget. The qualitative examples in Figure 1b indicate that this behavior is primarily a failure to terminate, because the student often produces a correct answer well before the end of the response and then continues with redundant or repetitive text. Table 1 provides an immediate clue. The Qwen3 base checkpoint declares <|endoftext|> as its EOS token, whereas the post-trained checkpoint also recognizes <|im_end|> as a conversational termination token (Qwen Team, 2025). The student and the teacher therefore share a tokenizer and a vocabulary, but their declared termination conventions differ. A first hypothesis is that the rollout decoder does not recognize all termination tokens that are relevant to this pair, in which case aligning the decoding stopping set would be sufficient.

Fix 1: Shared-set decoding. We first test the most direct correction at the decoding level. For the Qwen3 pair, we register both <|endoftext|> and <|im_end|> as valid stopping tokens during student rollout, while

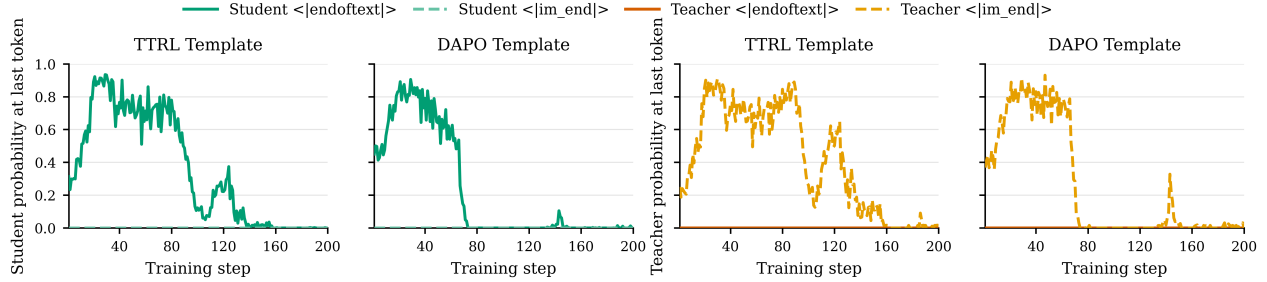


Figure 2 Qwen3 termination-token probabilities under vanilla OPD. Probabilities are measured at the final generation position of each rollout, which either ends naturally or is truncated at the generation budget. The student favors `<|endoftext|>`, while the teacher favors `<|im_end|>` at the same rollout-end prefixes.

the OPD objective and the token probabilities of both models remain unchanged. As shown in Figure 4a, Fix 1 follows nearly the same response-length and clipping trajectory as vanilla OPD and does not remove the failure. Aligning the decoding stopping set is therefore not sufficient, which leads us to examine how termination is represented in the distillation signal.

The failure of Fix 1 motivates an examination of the learned distributions rather than the decoding configuration alone. Figure 2 shows a much stronger mismatch at the probability level. At the same rollout-end prefixes, the Qwen3 base student places most of its termination probability on `<|endoftext|>`, whereas the post-trained teacher favors `<|im_end|>`. The student also assigns very small probability to the teacher-preferred `<|im_end|>` token, approximately 10^{-11} around terminal states. Registering this token as an additional stopping token therefore has little practical effect, because the student almost never samples it.

The mismatch also changes the supervision that reaches the termination token which the student *does* sample. For a termination token e generated by the student, sampled-token OPD uses

$$A_t(e) = \log \pi^E(e \mid s_t) - \log \pi_\theta(e \mid s_t).$$

This coefficient is negative whenever the teacher assigns less probability to that particular surface token than the student does, even if the teacher assigns substantial probability to terminating through another termination-equivalent token. In this case, sampled-token OPD suppresses the EOS token that the student sampled instead of recognizing that the teacher may express the same semantic decision to stop through a different token. For Qwen3, the student therefore receives a suppressive signal on its native `<|endoftext|>` token, and at the same time it receives little direct supervision on the teacher-preferred `<|im_end|>`, which is rarely sampled. Consistent with this mechanism, the probability that the student assigns to its native EOS falls from roughly 0.8 early in training to near zero in Figure 2, while response length and clipping increase.

The low student probability of the teacher-preferred EOS makes this asymmetry particularly severe under sampled-token OPD. At each visited prefix, direct teacher supervision is obtained through the token sampled by the student. An alternative EOS with negligible student probability therefore receives very few direct sampled updates, while the student’s native EOS continues to receive the suppressive signal described above. This makes reliable transfer of the teacher’s preferred termination form difficult at practical sampling budgets, reflecting a sampling-coverage limitation related to those studied in on-policy optimization (Mei et al., 2021; Agarwal et al., 2021).

Adding the teacher-preferred token to the decoding stopping set does not address this asymmetry: it neither changes the token-level distillation signal nor directly increases the probability of sampling that token. In Section 4, we examine a complementary case in which the pretrained student already assigns non-negligible probability to the teacher-preferred EOS, allowing vanilla OPD to transfer the surface termination form, although termination later deteriorates. Fix 1 therefore distinguishes decoding alignment from alignment of the distillation signal: tokens treated as equivalent stopping events by the decoder are still supervised as distinct actions by the objective.

Remark: sampling versus objective. The difficulty is not solely a consequence of single-token sampling. Under a softmax parameterization, the full-vocabulary local reverse-KL gradient is also weighted by the student’s token probabilities, so a teacher-preferred EOS with negligible student probability can receive only a weak recovery signal even when the entire vocabulary is evaluated. Full-vocabulary computation removes current-token sampling noise, but does not remove this probability weighting or reconcile the semantics of different termination tokens. Because full-vocabulary OPD incurs substantially higher computational cost, we do not evaluate it here and restrict our empirical analysis to sampled-token OPD.

3.2 Probability-Level Termination Alignment

The failure of Fix 1 indicates that the termination mismatch must be addressed in the distillation signal, so we consider corrections that reconcile the termination probabilities of the teacher and the student. Let $\mathcal{E}_{\text{EOS}}$ denote the termination-equivalent tokens under the rollout protocol of a pair, and let $e_\star \in \mathcal{E}_{\text{EOS}}$ be a canonical EOS token that the base student supports. Fix 1 modifies only the decoding interface. We next consider three ways of aligning the termination signal itself, which modify the teacher distribution, the semantic action used by the objective, and the student action space.

Fix 2: Teacher-side EOS mapping. We map the probability mass that the teacher places on all termination-equivalent tokens to the canonical student EOS token:

$$\tilde{\pi}^E(e_\star | s_t) = \sum_{e \in \mathcal{E}_{\text{EOS}}} \pi^E(e | s_t).$$

This idealized mapping sets the teacher probabilities of the other EOS tokens to zero. In practice we retain negligible probability on them for numerical stability and adjust the canonical probability so that the total mass is preserved. The student distribution is unchanged, and rollout terminates only on $e_\star$, which transfers the stopping signal of the teacher to a token that the student already supports.

Fix 3: Semantic EOS class. Rather than choosing a canonical surface token, we treat all tokens in $\mathcal{E}_{\text{EOS}}$ as realizations of a single semantic action STOP. For either $\pi \in \{\pi_\theta, \pi^E\}$, define

$$\bar{\pi}(\text{STOP} | s_t) = \sum_{e \in \mathcal{E}_{\text{EOS}}} \pi(e | s_t),$$

while $\bar{\pi}(a | s_t) = \pi(a | s_t)$ for non-EOS tokens. If the sampled token y_t is an EOS token, we replace it by $\bar{y}_t = \text{STOP}$, and otherwise $\bar{y}_t = y_t$. We then apply the standard sampled-token OPD update directly:

$$A_t = \log \bar{\pi}^E(\bar{y}_t | s_t) - \log \bar{\pi}_\theta(\bar{y}_t | s_t), \quad g_t = A_t \nabla_\theta \log \bar{\pi}_\theta(\bar{y}_t | s_t).$$

EOS samples are thus supervised through the total termination probability, non-EOS tokens are unchanged, and all tokens in $\mathcal{E}_{\text{EOS}}$ are registered as valid stopping tokens during rollout.

Fix 4: Canonical single-EOS action space. We map the EOS probability mass of the teacher to $e_\star$ as in Fix 2, and we remove all other tokens in $\mathcal{E}_{\text{EOS}}$ from the sampling distribution of the student, which we renormalize over the remaining probabilities. The same renormalized distribution is used for the actor log-probability computation, and rollout terminates only on $e_\star$. This produces a consistent action space with a single canonical termination action, irrespective of how many termination-equivalent tokens the original vocabulary contains.

The ablation in Figure 4a completes the diagnostic sequence in Section 3.1. Fix 1, which changes only the rollout stopping set, remains close to vanilla OPD, whereas Fixes 2, 3, and 4 reconcile termination at the probability level and behave similarly: response length and clipping remain substantially closer to the teacher references, and the native termination probability of the student no longer collapses. This comparison isolates the part of the procedure that matters, because adding `<|im_end|>` to the stopping set neither makes the initially unsupported token likely to be sampled nor changes the negative supervision that the frequently sampled `<|endoftext|>` receives. Only a correction of the termination signal at the probability level addresses the mismatch directly.

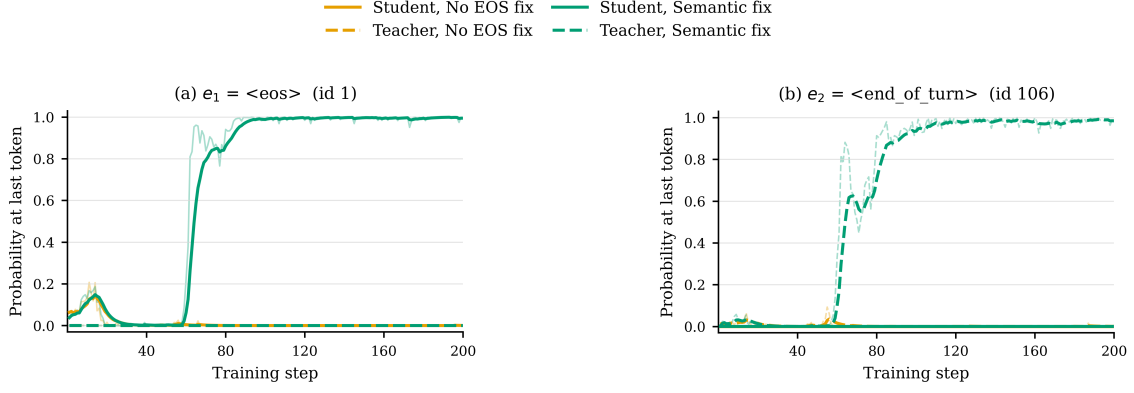
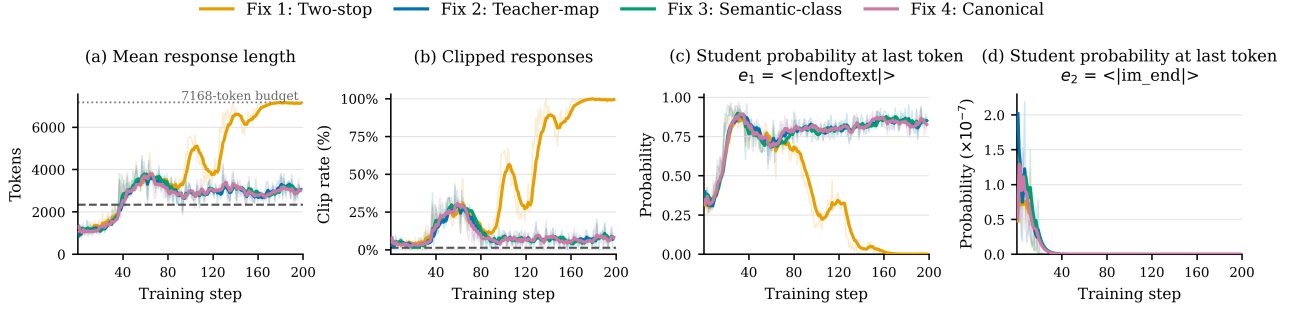
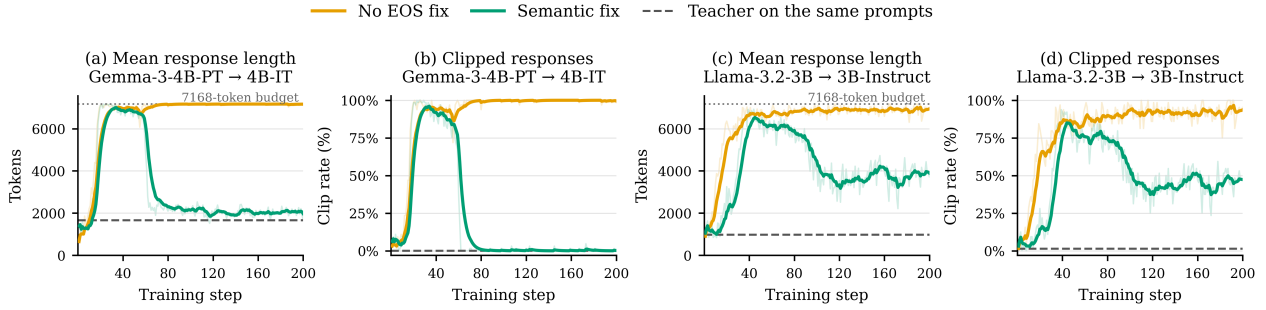


Figure 3 Gemma 3 termination preferences within a shared EOS set. Probabilities are measured at the final generation position of each rollout, which either ends naturally or is truncated at the generation budget. Despite sharing the same declared EOS set, the Gemma 3 PT student and IT teacher favor `<eos>` and `<end_of_turn>`, respectively. The semantic correction recovers termination without forcing these surface preferences to coincide.



(a) Qwen3: comparison of four EOS corrections.



(b) Gemma 3 (left) and Llama 3.2 (right): vanilla OPD versus semantic EOS correction.

Figure 4 EOS correction across model families. The semantic EOS correction mitigates length inflation in Qwen3 (top), Gemma 3 (bottom left), and Llama 3.2 (bottom right), with different recovery dynamics and residual gaps from the teachers. The Qwen3 ablation additionally compares teacher-side EOS mapping and canonical single-EOS mapping, which behave similarly to semantic aggregation, against two-stop decoding, which is insufficient. Gray reference lines in the length and clipping plots indicate the corresponding teacher statistics on the same prompts. All experiments use the TTRL prompt template.

Which correction to adopt. Fixes 2, 3, and 4 are empirically similar in the Qwen3 ablation, but they differ in the assumptions they require. For the Qwen3 pair studied above, teacher-side EOS mapping (Fix 2) is already sufficient and provides the simplest correction, because once the corresponding student and teacher termination tokens are known it modifies only the teacher distribution and leaves both the student distribution and the sampled-token OPD update unchanged, while canonical single-EOS decoding (Fix 4) further constrains the action space of the student. Fixes 2 and 4, however, both require a particular surface token to be designated

as the canonical representation of termination. This is natural for a known Qwen3 mismatch, but it is less convenient when the correction is extended to model families with different termination conventions, and especially when the student and the teacher expose the same declared EOS set but prefer different tokens within it. Semantic EOS aggregation (Fix 3) avoids this requirement, because it matches the total probability assigned to termination without designating a canonical surface form and without requiring the two models to learn identical token preferences. We therefore use Fix 2 as the simplest correction for the explicit Qwen3 mismatch, and we adopt semantic EOS aggregation as the default formulation for the cross-family experiments in [Section 3.3](#) (the same base-to-post-trained distillation repeated within Qwen3, Llama 3.2, and Gemma 3, not distillation between families).

Template and grader effects. The shared termination failure should be distinguished from its effect on measured task performance. The Qwen3 template comparison in [Figure 5](#) shows that cross-template evaluation can reduce the observed response length, most clearly when a DAPO-trained model is evaluated with the TTRL template, and that performance gains from OPD can persist despite severe length inflation, particularly under TTRL evaluation. The apparent effect on accuracy also depends on the grader, because DAPO-style evaluation is more sensitive to long redundant continuations ([Section B.3](#)). Length inflation therefore does not necessarily imply a comparable loss of reasoning capability. These observations about templates complement the cross-family termination mechanism rather than define it.

3.3 Termination Mismatch Across Model Families

The EOS configurations in [Table 1](#) illustrate why a representation-independent formulation is useful. Llama 3.2 resembles Qwen3, because its base checkpoint declares `<|end_of_text|>` while the post-trained checkpoint also recognizes conversational termination tokens such as `<|eom_id|>` and `<|eot_id|>`. Gemma 3 provides a complementary and more general case. Its PT and IT checkpoints already declare the same stopping tokens, `<eos>` and `<end_of_turn>`, so no token is missing from the decoding stopping set. [Figure 3](#) nevertheless shows that their learned preferences differ sharply: the student assigns its stopping probability mainly to `<eos>`, whereas the teacher favors `<end_of_turn>`, and with the semantic correction the student recovers reliable termination without adopting the surface token that the teacher prefers. The relevant mismatch is therefore more general than a discrepancy between declared EOS sets, since models can expose the same decoding interface while they distribute their learned termination probability differently across semantically equivalent tokens.

The cross-family results in [Figure 4](#) support this interpretation. Relative to vanilla OPD, the semantic EOS correction substantially reduces response length and clipping in Qwen3, Llama 3.2, and Gemma 3, where Qwen3 and Gemma 3 approach their corresponding teacher reference lengths relatively closely and Llama 3.2 retains a larger residual gap. The shared effect is therefore not that the student is forced to reproduce the EOS token that the teacher prefers. Semantic aggregation instead removes the competition between surface tokens by supervising their total probability as a common stopping action, and the different recovery trajectories indicate that correcting this mismatch does not imply identical length dynamics across model families. The mechanism further predicts that the termination action of the student is suppressed from the beginning of training when the teacher-preferred termination token lacks sampling support. The midtraining and SFT checkpoints of K2-Horizon provide the corresponding control, because they already assign substantial probability to the teacher-preferred termination token, and vanilla OPD from these checkpoints shows neither the termination collapse observed above nor unstable response length and clipping. We develop this contrast in [Section 4](#), where we also show that sampling support alone does not keep termination stable throughout training.

4 Going Beyond the EOS Mismatch

It was demonstrated in [Section 3](#) that aligning termination at the probability level removes the length inflation caused by the mismatch. This section addresses two questions that the mismatch does not answer: under which conditions the mismatch is harmful, and which length dynamics remain after the correction.

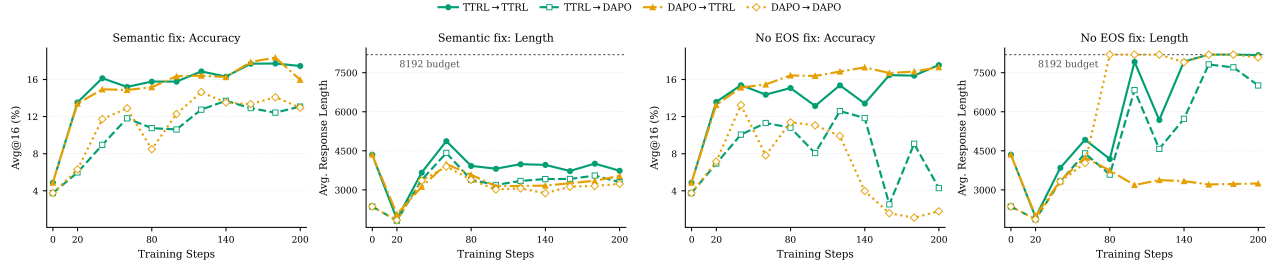


Figure 5 Template-dependent evaluation in Qwen3 with and without EOS correction. We distill Qwen3-4B into Qwen3-1.7B-Base on the DAPO dataset and report Avg@16 over AIME24, AIME25, and AMC23, computed using 16 sampled responses per problem. In the legend, $A \rightarrow B$ denotes that the student is trained with template A and evaluated with template B.

4.1 OPD Across Post-Training Stages

The experiments above compare base students with post-trained teachers. To examine how termination behavior depends on the student’s training stage, we use K2-Horizon 7B, which includes checkpoints spanning pretraining, midtraining, supervised fine tuning, and the final post-trained model. We fix the final checkpoint as the teacher and initialize the student from different earlier stages.

K2-Horizon provides a useful stage-wise comparison because the relevant termination-token IDs and the declared stopping set remain unchanged across stages (Table 1). We can therefore examine how learned termination preferences and OPD dynamics evolve over training without conflating them with changes in the declared EOS configuration.

Figure 7 shows a clear stage-dependent shift in termination preference. The pretrained student primarily favors `<|ifm|endoftext|>`, whereas the final post-trained teacher favors `<|ifm|im_end|>`. Most of this shift already occurs during midtraining: the midtraining checkpoint places substantial probability on the teacher-preferred termination token, and by the SFT stage the student’s termination preference is already close to that of the final teacher.

The Pretrain-to-Final run also provides a useful contrast with Qwen3. Although the pretrained K2-Horizon student and final teacher initially prefer different termination tokens, the student already assigns non-negligible probability to the teacher-preferred token. It can therefore sample this token and receive direct teacher supervision on it. Under vanilla OPD, termination mass consequently shifts toward the teacher-preferred surface form, showing that sampled-token OPD can transfer an alternative termination token when it already has sufficient support under the student policy.

Importantly, this transfer does not prevent a later stopping failure. After the teacher-preferred termination form has been learned, termination probability decreases again and response length grows toward the generation budget. Thus, surface-token alignment alone does not account for the full Pretrain-to-Final dynamics. We return to this late-stage behavior in Section 4.2.

A non-interference check at the midtraining and SFT stages. The midtraining and SFT checkpoints provide a control for the semantic correction. The midtraining and SFT students already place substantial probability on the teacher-preferred termination token, so vanilla OPD from these checkpoints does not exhibit the severe termination failure observed in the more mismatched base-to-post-trained settings. Applying semantic EOS aggregation in these cases produces similar response-length, clipping, and termination dynamics, with no consistent degradation in downstream performance. When the termination representations are already sufficiently aligned, the correction is therefore largely inert rather than a force that pushes the student toward different behavior. We provide the full training and evaluation comparisons in Section D.2.

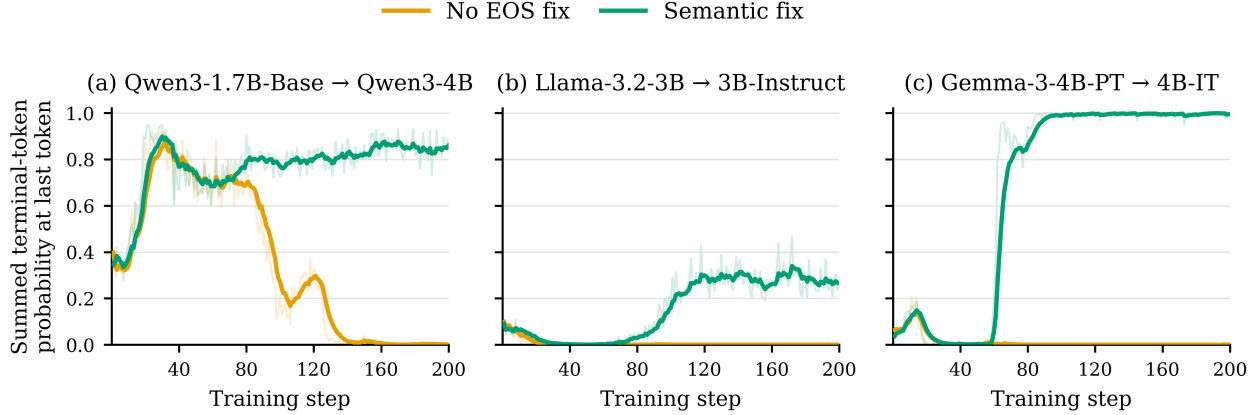


Figure 6 Evolution of termination probability across model families. Summed probability assigned by the student to its termination tokens at the final observed rollout position for (a) Qwen3, (b) Llama 3.2, and (c) Gemma 3. Under vanilla OPD, termination probability falls to near zero in all three settings. The semantic EOS correction prevents this collapse or enables recovery, but the dynamics differ across families: Qwen3 quickly reaches and maintains a high termination probability, whereas Llama 3.2 and Gemma 3 remain in a low-probability regime for longer and recover only later in training.

4.2 Residual Length Dynamics

Removing the surface EOS mismatch does not eliminate all response-length dynamics. We distinguish early length growth after correction from the renewed inflation observed late in the K2-Horizon run (Phase 3 in Figure 7).

Early length growth after EOS correction. Even with semantic EOS correction, response length initially increases before decreasing across Qwen3, Llama 3.2, and Gemma 3 (Figure 4). This transient drift is weak in Qwen3 and more pronounced in Llama 3.2 and Gemma 3. Student responses also remain longer than the corresponding teacher reference at later checkpoints, with the largest residual gap in Llama 3.2.

The termination dynamics in Figure 6 are consistent with these differences. With semantic correction, Qwen3 recovers its stopping probability relatively quickly, whereas Llama 3.2 and Gemma 3 remain in a low-termination regime for longer. These curves describe different recovery dynamics, but do not by themselves establish their cause.

The corrected K2-Horizon Pretrain-to-Final run also exhibits an initial increase in response length, marked as Phase 1 in Figure 7. Unlike the subsequent recovery seen in the other families, this increase is followed by an elevated, fluctuating plateau in Phase 2. The plateau should therefore not be interpreted as recovery to the teacher’s response length or as convergence of the termination distributions.

In Section C, we analyze a possible contributor to early length growth from the trajectory-level reverse KL perspective. Local sampled-token OPD omits the future student–teacher mismatch incurred by continuation. At a fixed prefix, this omission removes a nonnegative contribution to the stopping-logit update relative to the trajectory-consistent reverse KL update. The resulting relative continuation bias is consistent with early length growth after EOS correction, including Phase 1 of the corrected K2-Horizon run.

Late-stage re-inflation in K2-Horizon. Phase 3 of the Pretrain-to-Final runs reveals an additional failure. After the intermediate plateau, response length increases again, termination probabilities approach zero, and almost all rollouts eventually reach the generation budget. This transition is relatively abrupt under vanilla OPD and more gradual under semantic EOS correction.

In the vanilla run, the renewed inflation occurs after naturally terminated rollouts have already shifted toward the teacher-preferred EOS token. More importantly, the corrected run exhibits the same qualitative late-stage

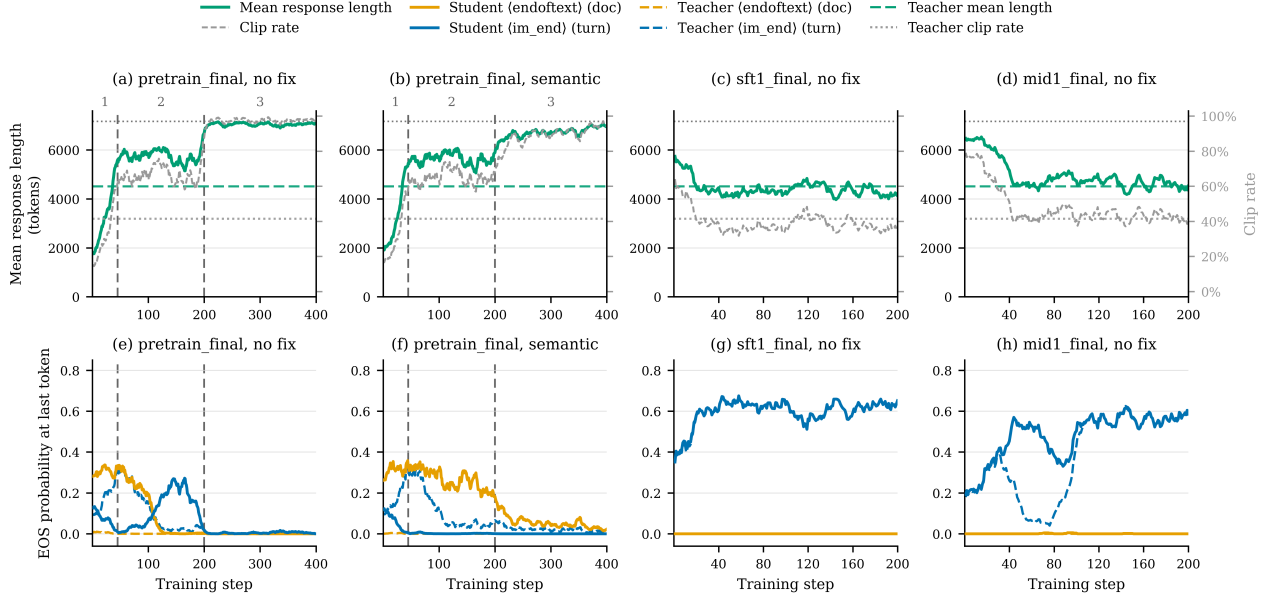


Figure 7 OPD from different K2-Horizon training stages to the final post-trained model. The top row shows response length and clipping rate, and the bottom row shows student and teacher probabilities on the two termination tokens at the final rollout position. We compare Pretrain-to-Final distillation with and without semantic EOS correction, together with vanilla OPD initialized from the SFT and midtraining checkpoints. Horizontal green dashed and gray dotted lines denote the teacher mean length and clipping rate, respectively. Labels 1–3 mark the initial growth, intermediate plateau, and late-stage re-inflation phases of the Pretrain-to-Final runs; vertical gray dashed lines indicate their approximate boundaries.

behavior even though the objective already treats both EOS tokens as a shared stopping action. Surface termination-token identity alone therefore cannot explain this renewed collapse.

The late-stage increase is qualitatively similar to the length inflation and truncation collapse reported by Luo et al. (2026), but this similarity does not establish a shared mechanism. We leave the origin of this late-stage K2-Horizon collapse, together with the differences in recovery dynamics across model families, for future work.

5 Related Works

Length inflation in on-policy distillation. On-policy distillation trains the student on its own sampled rollouts under per-token supervision from a teacher (Lu and Lab, 2025; Yang et al., 2025). Uncontrolled length growth during OPD has been reported in several recent works, each identifying a distinct mechanism. Luo et al. (2026) observe abrupt length inflation accompanied by repetition saturation, and attribute it to an implicit preference of the distillation objective for long and repetitive rollouts. Li et al. (2026) find that overly large generation budgets lead to late-stage training collapse, and attribute it to the degradation of teacher supervision quality with trajectory depth. Zhao et al. (2026) relate length inflation to entropy collapse induced by reverse-KL distillation. These mechanisms are not mutually exclusive, and each may well be operative in the settings where it was identified. They do, however, share a precondition: the student must first fail to terminate. In this work we revisit that precondition across base–post-trained model pairs from several widely used model families, and find that the two models frequently encode termination through different special tokens. Where this is the case, the token-level objective penalizes the student’s stopping action from the outset, producing the same truncation-and-repetition signature studied above but through a cause that precedes those mechanisms and is removed by a change to the objective alone. Our results are therefore complementary to prior analyses rather than in tension with them: in settings where teacher and student already agree on termination, the mechanisms above remain the relevant explanation, and where they do not, termination alignment should be established before length inflation is attributed to an algorithmic cause.

Rollout quality degradation and its mitigation. Beyond length inflation, a second line of work focuses on the degradation of rollout quality itself. Once the student errs or begins repeating midway through a trajectory, its subsequent prefixes drift away from the teacher’s normal distribution; the teacher’s preferences over these degenerate prefixes become decoupled from task correctness, the distillation signal grows increasingly misaligned, and this in turn pushes the student toward even more degenerate generations. Liu et al. (2026) characterize this as *local teachability collapse*, and Xie et al. (2026) show that late-trajectory tokens systematically drift off the teacher distribution. Two families of remedies have been proposed. The first distills only on the early segment of a trajectory, where teacher supervision remains reliable, as in Prefix OPD (Zhang et al., 2026), Prune-OPD (Yang et al., 2026b), and ADWIN (Liang et al., 2026), which adapts the rollout horizon according to prefix drift. The second improves rollout quality directly so that the student avoids entering degenerate segments, as in SWITCH (Koo et al., 2025), which injects teacher tokens where the teacher and student disagree, the teacher-blended behavior policy of Plyusov et al. (2026), and Relay-OPD (Xu et al., 2026), which lets the teacher take over at the point where the student’s reasoning first goes wrong. All of these works treat rollout degradation as an algorithmic pathology arising from the interaction between the distillation objective and on-policy sampling. We instead show that the student’s inability to stop as training proceeds has a lower-level, implementation-level root cause, namely a termination-token mismatch, which alone suffices to produce budget saturation without any trajectory-level pathology.

Termination-token semantic mismatch. The Qwen3 base model declares `<|endoftext|>` as its termination token, whereas the post-trained model uses `<|im_end|>` as its primary conversational termination token while retaining `<|endoftext|>` as an additional stopping token (Qwen Team, 2025).

Prior work has identified related special-token issues in OPD. Fu et al. (2026) identify tokenizer and special-token mismatch, including EOS markers, as a failure mode of sampled-token supervision and mitigate it by masking problematic special tokens from the distillation signal. Plyusov et al. (2026) address the Qwen3 EOS mismatch through EOS canonicalization, mapping the native student and teacher EOS tokens to a shared stopping event before sampling and KL evaluation.

Our work instead takes termination mismatch itself as the object of study. We systematically trace how token-level OPD can amplify a disagreement between surface termination tokens into termination-probability collapse and length inflation, and compare multiple interventions that separately modify the decoding rule, teacher distribution, semantic action space, and student action space. This comparison shows that aligning the decoding stopping set alone is insufficient, whereas aligning termination at the probability level consistently mitigates the failure.

Moreover, we show that the phenomenon is not specific to Qwen3 or to an explicit difference in declared EOS configurations. Across Qwen3, Llama, and Gemma, student and teacher models can prefer different surface tokens for the same semantic stopping action; in Gemma 3, this mismatch arises even when the two models already declare the same EOS set. These results motivate treating termination as a semantic event rather than relying solely on model-level EOS configuration.

6 Limitations and Future Work

Our experiments across Qwen3, Llama, and Gemma identify termination-token mismatch as an important source of length inflation, but not a complete explanation of the phenomenon. The mismatch can arise either from different declared stopping sets or from different learned preferences among termination tokens even when the declared sets coincide. Our K2-Horizon experiments further show that, after the student has successfully acquired the teacher-preferred termination token, the probabilities of both termination tokens can decrease again later in training, accompanied by a second increase in response length; a similar late-stage increase remains under semantic EOS correction. We therefore do not attribute these residual dynamics to termination-token mismatch and leave their underlying mechanism for future work. Consistent with this limitation, prior work also reports length inflation in settings without the same declared stopping-set mismatch (Luo et al., 2026). Characterizing these remaining mechanisms and the conditions under which they dominate is an important direction for future study.

A second limitation concerns the extension beyond single-turn tasks. Our experiments consider single-turn mathematical reasoning, where the tokens grouped into the semantic EOS class are operationally equivalent under the decoding protocol: each ends the response. This equivalence need not hold in multi-turn or agentic settings, where special tokens may distinguish message completion, the end of an assistant turn, document termination, and a handoff to a tool. The base-to-post-trained pairing used here would also not carry over: most base models lack tool-use and multi-turn capability, so a fair comparison would instead require two post-trained models of different sizes, which is a substantially different experimental design. Treating these functionally distinct events as interchangeable alignment targets could alter interaction control flow rather than merely correct stopping behavior. Extending our approach therefore requires context-dependent termination classes that preserve the semantics of the underlying interaction protocol. Length inflation may also have more severe consequences in these settings: repetition can exhaust the context budget or disrupt subsequent tool calls, affecting task success rather than token count alone. These effects are not evaluated in our current experiments.

Finally, our prompt-template analysis is limited to the Qwen3 experiments with the three templates and two graders studied here. Within this scope, the severity of length inflation depends strongly on the training and evaluation templates, while the associated downstream performance gains can transfer across templates. The cross-family evidence for termination mismatch does not, by itself, establish that these template effects generalize to Llama or Gemma. A more systematic analysis across model families, task domains, and evaluation protocols is needed to distinguish template-specific generation artifacts from more general changes in reasoning capability.

7 Conclusion

We document and analyze length inflation in on-policy distillation. One major cause is a termination mismatch: base students and post-trained teachers mark the end of a response with different special tokens. The token-level distillation objective therefore treats the student’s own termination as an over-produced token and suppresses it, so the student’s termination probability decays over training and responses keep growing. We compared four fixes and found that aligning the two termination tokens into a single stop action at the probability level restores response length and clipping ratio to the teacher’s reference levels, whereas merely enlarging the decoding stop set does not. This confirms from the opposite direction that the problem lies in the training objective rather than the decoding setup. Our template experiments further show that the training template has a persistent effect on response length, while the evaluation format substantially changes measured accuracy, and the two need not move together. Overall, termination semantics is an overlooked but important axis of alignment in on-policy distillation, and future work will extend to multi-turn and agentic settings and examine other mechanisms behind length inflation.

Author Contributions and Acknowledgments

Four student authors (YY, TY, SL, KZ) contribute equally to this project: The length inflation is observed by SL, SL proposed several mechanisms and solutions to mitigate this. However, the actual EOS mismatch is identified by YY in adjacent projects supervised by WZ in 2026 summer. YY verified this with SL on Qwen. SL, TY and KZ extends this to Gemma, Llama and check the staged phenomena in K2-Horizon.

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A Experimental Details

Implementation and compute. Our implementation is built on the public OPD codebase of [Li et al. \(2026\)](#), with modifications for the EOS corrections and evaluation protocols studied in this work. Unless otherwise specified, all experiments can be run on four NVIDIA RTX PRO 6000 GPUs with 96 GB of memory per GPU. We will release the code and experiment configurations to facilitate reproduction.

Main setup. Unless otherwise specified, the main experiments in this note use Qwen3-1.7B-Base as the student and Qwen3-4B as the teacher, trained on DAPO-Math-17K ([Yu et al., 2026](#)) with thinking disabled. The default setting uses the TTRL format for both training and evaluation. The exact prompt templates are provided in [Section B.1](#). The teacher remains fixed throughout training.

We use sampled token OPD. Student responses are generated on policy, and the distillation signal at each generated token is given by the teacher and student log probability difference on that token. Each training step contains 16 prompts with four sampled responses per prompt, corresponding to 64 trajectories per step. We train for 200 steps and save checkpoints every 20 steps.

EOS correction experiments. The EOS correction experiments use the same training configuration and differ only in their treatment of termination actions. We compare vanilla OPD, two stop decoding, teacher side EOS mapping, semantic EOS aggregation, and canonical single EOS mapping, as described in [Section 3.2](#).

Checkpoint evaluation. We independently evaluate checkpoints from steps 20 through 200 on AIME24, AIME25, and AMC23. These benchmarks contain 30, 30, and 40 problems, respectively. Each problem is sampled 16 times. Evaluation uses a fixed sampling seed of 0 for all methods and checkpoints. Unless otherwise specified, evaluation uses the TTRL format with thinking disabled.

Table 2 Training hyperparameters used in the main experiments.

Hyperparameter	Value
Rollout temperature	1.0
Rollout top- p	1.0
Rollout top- k	None
Optimizer	AdamW
Learning rate	1×10^{-6}
Learning rate schedule	Constant
Adam betas	(0.9, 0.999)
Weight decay	0.01
Gradient clipping	1.0
Optimization epochs per batch	1
Mini batch size	16
Loss aggregation	Token mean

Table 3 Sampling hyperparameters used for checkpoint evaluation.

Hyperparameter	Value
Samples per problem	16
Temperature	0.7
Top- p	0.95
Top- k	None
Repetition penalty	1.0

Evaluation metrics. For each problem, we compute the mean correctness over its 16 sampled responses. We first average over problems within each benchmark and then report the unweighted macro average over AIME24, AIME25, and AMC23 as EOS termination rate, format compliance, and answer extraction success when relevant.

Generation lengths. For the Qwen3, Llama 3.2, and Gemma 3 experiments, training uses a maximum prompt length of 1,024 tokens and a maximum response length of 7,168 tokens, with a maximum model context of 8,192 tokens. Independent checkpoint evaluation allows up to 8,192 generated tokens with a maximum model context of 12,288 tokens.

K2-Horizon stage-wise experiments. For the K2-Horizon experiments, training uses a maximum prompt length of 1,024 tokens and a maximum response length of 7,168 tokens, with a maximum model context of 8,192 tokens, matching the native context length of the pretrained checkpoint. Checkpoint evaluation allows up to 7,168 generated tokens. The Pretrain-to-Final runs are trained for 400 steps, whereas the Midtrain-to-Final and SFT-to-Final runs are trained for 200 steps. Unless otherwise specified, the remaining OPD configuration follows the main experimental setup.

B Prompt and Evaluation Protocols

B.1 Prompt Templates

We consider three prompt templates for generating student rollouts. The exact templates used in our experiments are provided below. The placeholder `{problem}` is replaced by the corresponding problem statement.

DAPO Template

Solve the following math problem step by step. The last line of your response should be of the form Answer: \$Answer (without quotes) where \$Answer is the

answer to the problem.

{problem}

Remember to put your answer on its own line after "Answer:".

TTRL Template

{problem} Please reason step by step, and put your final answer within `\boxed{}`.

Raw-Question Template

{problem}

B.2 Evaluation Graders

We evaluate responses under two answer formats corresponding to the DAPO and TTRL templates. The raw-question template is used only for training, as it does not prescribe a canonical final-answer format for reliable automatic extraction. For TTRL-style evaluation, we extract the final `\boxed{}` answer. For DAPO-style evaluation, we start from the official DAPO grader, which extracts the answer following the final **Answer:** marker within the last 300 characters of the response and applies normalized exact matching.

We find, however, that the unmodified DAPO grader substantially underestimates the performance of Qwen3 models because it is sensitive to equivalent Markdown layouts of the final answer. As shown in Figure 8, the Qwen3-4B teacher obtains only 6.28% Avg@16 under the original grader, compared with 24.34% after applying our format-compatible parsing extension. Under the original grader, several student checkpoints even appear to outperform the teacher, despite the large discrepancy being primarily attributable to answer formatting rather than mathematical correctness.

Manual inspection shows that Qwen3 frequently places the answer after **Answer:** using display-math formatting, including multiline `$$...$$` blocks, which can be missed by the original line-based extraction. We therefore use a minor Qwen3-compatible extension of the DAPO grader throughout our DAPO-format evaluations. The extension changes only answer parsing; the original DAPO normalization and exact-match criterion are retained. Details are provided in Section B.3.

Why answer format matters more in OPD. This issue is particularly relevant when evaluation protocols designed for RLVR are reused for OPD. In RLVR, a format sensitive verifier or reward provides direct training pressure toward outputs that can be recognized by the corresponding grader. In OPD, however, no such outcome reward is present. The student’s output format is instead determined mainly by the teacher distribution together with the prompt. Consequently, a grader that assumes a particular answer layout may underestimate performance when that layout differs from the format naturally preferred by the teacher. This motivates using an evaluation parser that preserves the original correctness criterion while accommodating equivalent answer layouts.

B.3 DAPO-Compatible Evaluation Grader

The official DAPO grader considers only the final 300 characters of a response, extracts the answer following the last case-insensitive **Answer:** marker, applies the official DAPO normalization to the prediction and ground truth, and performs normalized exact matching.

Qwen3 frequently renders valid answers using display-math formatting, for example

Answer: `$$\boxed{204}$$`,

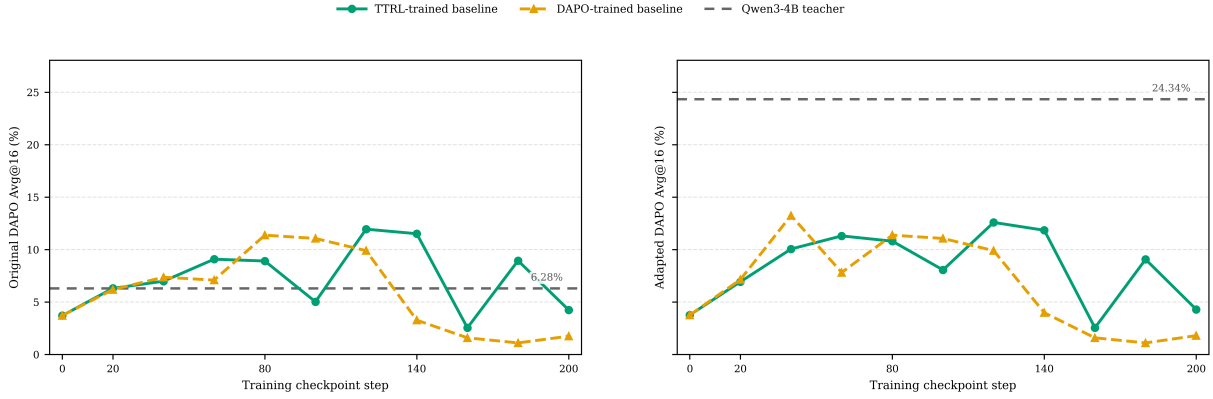


Figure 8 Effect of DAPO answer parsing on Qwen3 evaluation. We compare the original DAPO grader (left) with our format-compatible extension (right) for vanilla OPD using Qwen3-1.7B-Base as the student and Qwen3-4B as the teacher. The dashed horizontal line denotes teacher performance under the corresponding grader. The teacher score increases from 6.28% to 24.34% Avg@16 when equivalent Qwen3 LaTeX answer layouts are parsed correctly, indicating that the original grader can substantially underestimate Qwen3 performance due to formatting.

or with the display-math block beginning on the following line. Such outputs can be misparsed by the original line-based extraction.

We therefore first apply the unmodified official DAPO scorer and invoke a fallback parser only when it returns an incorrect result. The fallback uses the same final-300-character window and still requires the answer to follow the final **Answer:** marker, but additionally supports equivalent inline and display LaTeX layouts, including multiline $\\$\\$\\$\\$$ blocks. After extraction, we apply exactly the same DAPO normalization and exact-match comparison as the official grader. No semantic matching, symbolic equivalence, or search for answers elsewhere in the response is introduced.

C Future Mismatch and Transient Length Drift

After correcting the termination-token mismatch, we no longer observe the persistent budget saturation seen in vanilla OPD. However, response length can still increase during the early stage of training and decrease again later, a pattern that is particularly pronounced in our Llama and Gemma experiments.

Empirically, at early steps we often observe low termination probabilities near the end of a rollout, with the teacher assigning even less probability to stopping than the student at many such prefixes. At these prefixes, the local OPD coefficient for a sampled stopping action is negative. This does not by itself determine the expected stopping-logit update, which also depends on the coefficients assigned to continuation tokens. Moreover, the local signal does not account for the mismatch encountered after continuing along the student’s future trajectory.

This distinction is closely related to MiniLLM (Gu et al., 2024), which studies sequence-level reverse KL distillation through policy gradient optimization and decomposes the gradient into single-step and long-range contributions. Here, we use the unnormalized trajectory-level gradient as an analytical reference for local sampled-token OPD, rather than identifying it with MiniLLM’s final length-normalized update. We characterize a successor-KL contribution omitted by the local update and show that its stopping-logit component is nonnegative at a fixed prefix.

For this analysis, assume that the teacher is fixed and that both policies are defined on the same action space, with aligned termination semantics represented by a single stopping action e . We use a common finite generation budget H and assume common support and finite KL divergences. At position t , let $s_t = (x, y_{<t})$, whose prefix length determines the remaining budget, and let $f(s, y)$ denote the next state after predicting token y . Both predicting e and exhausting the generation budget lead to an absorbing terminal state $\dagger$. We analyze expected on-policy, unclipped score-function updates with trajectory-summed losses, before optimizer preconditioning;

the comparison need not hold exactly for updates with stale rollouts, clipping, or length-dependent reweighting.

Define the local reverse KL cost

$$\rho_\theta(s, y) = \log \frac{\pi_\theta(y | s)}{\pi^E(y | s)},$$

and let

$$V_\theta(s) = D_{\text{KL}}(P_{\pi_\theta}^s \| P_{\pi^E}^s)$$

denote the reverse KL between the student and teacher distributions over the remaining sampled tokens up to stopping or budget exhaustion, with $V_\theta(\dagger) = 0$. By the chain rule of KL,

$$V_\theta(s) = \mathbb{E}_{y \sim \pi_\theta(\cdot | s)} [\rho_\theta(s, y) + V_\theta(f(s, y))]. \quad (1)$$

Local OPD omits the future mismatch. Let $J_x(\theta) = V_\theta(s_1)$ be the trajectory-level reverse KL for prompt x . For a student trajectory of length $T \leq H$, its negative gradient can be written as

$$-\nabla_\theta J_x(\theta) = \mathbb{E}_{\tau \sim P_{\pi_\theta}^{s_1}} \left[\sum_{t=1}^T A_t^* \nabla_\theta \log \pi_\theta(y_t | s_t) \right],$$

where

$$A_t^* = \log \frac{\pi^E(y_t | s_t)}{\pi_\theta(y_t | s_t)} - V_\theta(f(s_t, y_t)). \quad (2)$$

Here and below, the scalar coefficient multiplying the policy score is treated as detached in the corresponding score-function update. These are uncentered score-function coefficients; adding a state-only baseline does not change the expected update.

In contrast, sampled-token OPD uses the local signal defined in the preliminaries,

$$A_t = \log \frac{\pi^E(y_t | s_t)}{\pi_\theta(y_t | s_t)},$$

and therefore drops the successor value $V_\theta(f(s_t, y_t))$. This omission arises from local credit assignment rather than single-token sampling: differentiating the full-vocabulary local reverse KL at detached student-generated prefixes also omits the effect of the current action on future visited prefixes.

Importantly, the omitted term is zero when $y_t = e$, since termination transitions directly to the absorbing state. Nevertheless, it contributes to the stopping-logit signal through softmax coupling with continuation tokens.

Effect on the stopping logit. The difference between the two updates is particularly simple for the stopping action. Let

$$p_\theta(s) = \pi_\theta(e | s), \quad C_\theta(s) = \mathbb{E}_{y \sim \pi_\theta(\cdot | s)} [V_\theta(f(s, y))].$$

Since $V_\theta(f(s, e)) = 0$, $C_\theta(s)$ contains only the successor KL associated with continuation actions. We can further have the following proposition:

Proposition 1 (Missing stopping pressure in local OPD). *Let $\Delta_e^*(s)$ and $\Delta_e^{\text{OPD}}(s)$ denote the expected ascent updates to the stopping logit $z_\theta(s, e)$ under the trajectory-consistent and local OPD coefficients, respectively. At any fixed prefix s ,*

$$\Delta_e^*(s) - \Delta_e^{\text{OPD}}(s) = p_\theta(s)C_\theta(s) \geq 0.$$

Thus, local sampled token OPD always applies weakly less upward pressure to the stopping logit than the trajectory-consistent reverse KL update.

Proof. For any action b , softmax coupling gives

$$\frac{\partial \log \pi_\theta(y | s)}{\partial z_\theta(s, b)} = \mathbf{1}\{y = b\} - \pi_\theta(b | s).$$

Since the trajectory-consistent coefficient differs from local OPD by $-V_\theta(f(s, y))$, their expected update difference on $z_\theta(s, b)$ is

$$\pi_\theta(b \mid s) [C_\theta(s) - V_\theta(f(s, b))],$$

where

$$C_\theta(s) = \mathbb{E}_{y \sim \pi_\theta(\cdot \mid s)} [V_\theta(f(s, y))].$$

For the stopping action e , $f(s, e) = \dagger$ and $V_\theta(\dagger) = 0$. Therefore,

$$\Delta_e^*(s) - \Delta_e^{\text{OPD}}(s) = p_\theta(s)C_\theta(s) \geq 0.$$

□

Interpretation. [Proposition 1](#) describes a relative bias rather than an absolute suppression of stopping. Local OPD may still increase the stopping logit when the immediate teacher signal favors termination. The proposition instead states that, compared with the trajectory consistent reverse KL update, local OPD always misses the nonnegative stopping signal

$$p_\theta(s)C_\theta(s).$$

The source of this gap is the continuation actions rather than the stopping action itself. After stopping, the successor KL is zero. For a continuation action, however, the trajectory level objective additionally subtracts the future mismatch $V_\theta(f(s, a))$. Through softmax coupling, penalizing these continuation actions also increases the relative preference for stopping. Local sampled token OPD omits these successor penalties and therefore overweights continuation relative to the trajectory consistent update.

This difference is naturally largest when the student still differs substantially from the teacher over future continuations. From [equation 1](#),

$$V_\theta(s) = k_\theta(s) + C_\theta(s), \quad 0 \leq C_\theta(s) \leq V_\theta(s),$$

so that

$$V_\theta(s) \rightarrow 0 \implies p_\theta(s)C_\theta(s) \rightarrow 0.$$

Hence, as distillation reduces the future student teacher mismatch, the local and trajectory consistent updates become increasingly similar.

This provides a possible explanation for the transient length drift observed after correcting the termination token mismatch. As shown in [Figure 6](#), stopping probabilities are initially low and recover later in training, while response lengths first increase and then decrease. The effect is especially pronounced for Llama 3.2 and Gemma 3, where the low termination regime persists for longer. We therefore view the omitted future mismatch as a plausible contributor to these transient dynamics, rather than as a guarantee that local OPD must increase response length.

This mechanism is distinct from the termination token mismatch in [Section 3.1](#), which explains the persistent budget saturation observed in vanilla Qwen3 OPD.

D Additional Experimental Results

D.1 Termination Mismatch and Residual Length Dynamics Across Model Families

We report additional evaluation results for distillation with Gemma-3-4B, Llama-3.2-3B, and K2-Horizon-7B pretrain_final in [Figure 9](#). Under the TTRL evaluation template, the semantic EOS fix yields modest, model-dependent changes in single-turn mathematical reasoning performance. For Llama, both variants remain near zero throughout training. Together, these results indicate that improvements in termination behavior do not necessarily translate into consistent gains in mathematical reasoning performance.

To better understand the cross family length behavior, we examine how the student and teacher distribute



Figure 9 Evaluation results with and without the semantic EOS fix under the TTRL evaluation template for Gemma-3-4B, Llama-3.2-3B, and K2-Horizon-7B pretrain_final distillation. Columns show mean scores over 16 samples per problem on AIME24, AIME25, and AMC23, followed by their macro average.

probability across their individual termination tokens in Figure 10. For Gemma 3 and Llama 3.2, vanilla OPD exhibits a clear termination preference mismatch. In Gemma 3, the student primarily uses `<eos>`, whereas the teacher strongly prefers `<end_of_turn>`. Similarly, in Llama 3.2, the student primarily places termination mass on `<|end_of_text|>`, while the teacher prefers `<|eot_id|>`. Consequently, vanilla OPD suppresses the termination token naturally used by the student without reliably transferring the teacher’s preferred surface form. The semantic correction avoids this competition by matching their total probability of termination while allowing the two models to retain different surface token preferences.

K2 Horizon exhibits a different two-stage behavior. Early in training, the teacher and student prefer different termination tokens, but unlike Qwen3, Llama, and Gemma, the student’s probability on the teacher-preferred token is not vanishingly small. The token can therefore still be sampled with non-negligible probability, allowing sampled-token OPD to receive direct teacher supervision on it and gradually transfer termination mass toward the teacher-preferred surface form.

Later in training, however, the probabilities of both termination tokens decrease again. This second-stage decline cannot be explained by the surface EOS mismatch alone, since the teacher-preferred token has already been learned. We hypothesize that it is instead related to the local nature of sampled-token OPD: when the teacher assigns low termination probability at the current student-generated prefix, the local update can continue to favor continuation while ignoring the future student-teacher mismatch induced by doing so. We analyze this possible mechanism in Section C.

This suggests that termination token identity alone does not explain all of the observed length dynamics. At

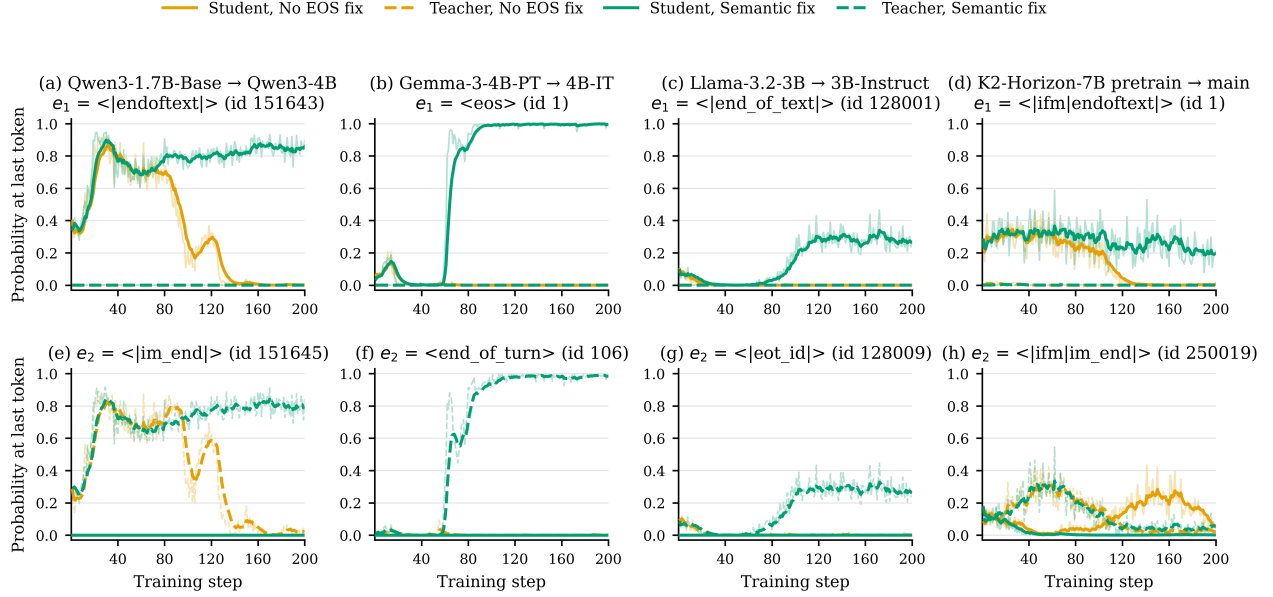


Figure 10 Evolution of individual termination token probabilities. We track the probabilities assigned by the student and teacher to each termination token at the final position of student rollouts for Qwen3 1.7B, Gemma 3 4B, Llama 3.2 3B, and K2 Horizon 7B. Each column is one student–teacher pair; the top row shows the document-level terminator e_1 and the bottom row the turn-level terminator e_2 . Llama 3.2 defines a third terminator, $\langle |eom_id| \rangle$, which we omit: its probability stays below 0.02 throughout training for both the student and the teacher under both settings.

prefixes where the teacher itself assigns little total probability to termination, sampled token OPD can still favor continuation even after the semantic mismatch is corrected. We study this additional effect in [Section C](#), where we analyze how the local OPD update differs from the trajectory level reverse KL update by ignoring future student teacher mismatch.

D.2 Semantic EOS Correction at Later Training Stages

The K2-Horizon experiments in [Section 4](#) show that the midtraining and SFT checkpoints already assign substantial probability to the teacher-preferred termination token. Unlike the pretrained initialization, vanilla OPD from these later-stage checkpoints does not exhibit the severe termination collapse observed in the base-to-post-trained settings studied in the main text. In these cases, semantic EOS correction is therefore not needed to recover the teacher’s preferred termination form.

We additionally test whether applying the correction when the termination mismatch is already substantially reduced perturbs either the training dynamics or downstream performance. This serves as a non-interference check for using semantic EOS aggregation as a general termination-handling rule.

As shown in [Figure 11](#), the semantic correction has little effect on the later-stage K2 training trajectories. For both the SFT and midtraining initializations, the student already places substantial mass on $\langle |ifm|im_end| \rangle$, and its termination behavior remains stable under vanilla OPD. Applying semantic aggregation leaves the response-length and clipping trajectories broadly unchanged and produces similar evolution of the termination probabilities. Thus, when the student and teacher are already sufficiently aligned at the level of termination, the correction is largely inactive rather than forcing a different surface termination convention.

The downstream results in [Figure 12](#) show the same pattern. For the SFT and midtraining initializations, the semantic and uncorrected curves remain close across AIME24, AIME25, AMC23, and their average, with small checkpoint-level differences in both directions. We do not observe a systematic accuracy penalty from aggregating the termination-equivalent tokens. The Pretrain-to-Final evaluation, included for completeness,

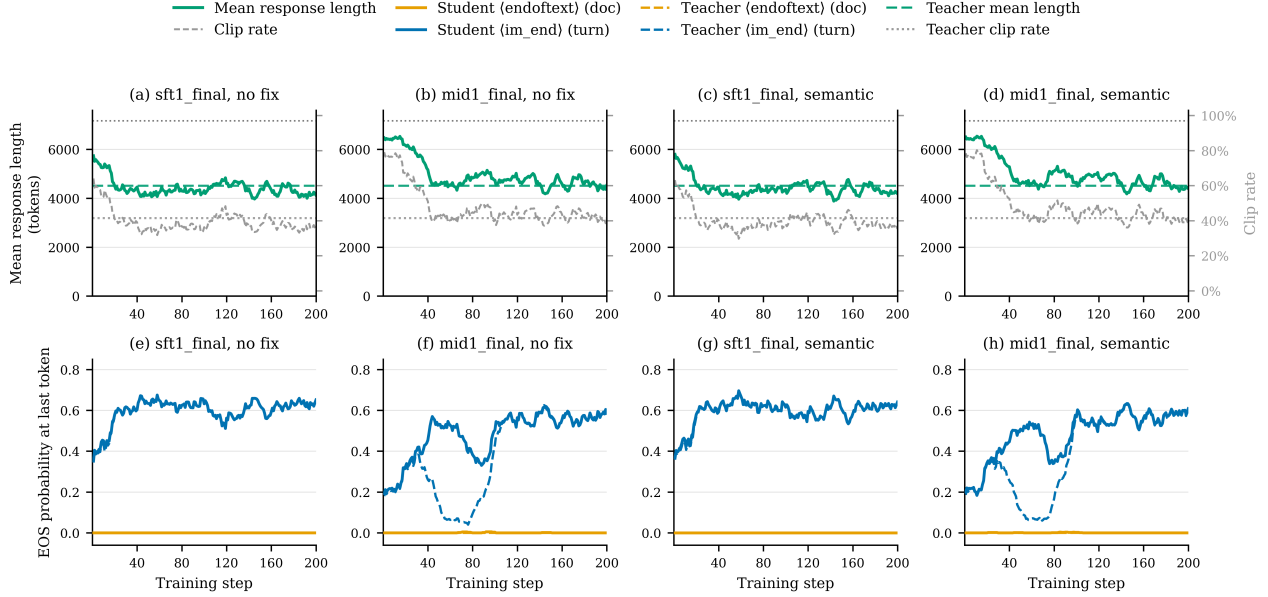


Figure 11 Semantic EOS correction at later K2-Horizon training stages. We distill the final post-trained K2-Horizon model into students initialized from the SFT and midtraining checkpoints, comparing vanilla OPD with semantic EOS correction. The top row shows mean response length and clipping rate together with the corresponding teacher reference levels, while the bottom row shows the student and teacher probabilities assigned to $\langle \text{im_end} | \text{endof_text} \rangle$ and $\langle \text{im_end} | \text{im_end} \rangle$ at the final observed rollout position. In both initialization settings, the corrected and uncorrected runs exhibit similar length, clipping, and termination-probability dynamics. In particular, semantic aggregation does not introduce the termination failure that it is designed to prevent in more severely mismatched settings.

similarly shows broadly comparable performance between the two variants despite their different treatment of termination.

Together, these results provide a useful control for the semantic EOS correction. Its main benefit arises in settings where surface termination preferences are strongly mismatched and sampled-token OPD cannot reliably transfer probability to the teacher-preferred token. When the student already provides substantial support for that termination form, as in the later K2-Horizon checkpoints, semantic aggregation leaves both the training dynamics and downstream performance largely unchanged in our experiments. This supports its use as a general correction without requiring that every teacher-student pair exhibit severe termination mismatch.

E Influence of Training and Evaluation Templates

Recall that Figure 5 suggests that changing the evaluation format relative to the training prompt can substantially alter the observed response length. We therefore study this interaction more systematically. Since Figure 4a shows that the EOS correction methods exhibit broadly similar behavior, we use the semantic EOS class as the default correction in this study.

Following the setup described above, we train Qwen3-1.7B-Base with Qwen3-4B as the teacher using each of the three training prompt templates, and evaluate every checkpoint under both the TTRL and DAPO formats. All other experimental settings are kept fixed.

The results are shown in Figure 13. We make three main observations.

Obs 1: The evaluation format strongly affects measured performance. Evaluation under the TTRL format generally yields higher measured accuracy than evaluation under the DAPO format across all three training settings. Matching the evaluation format to the training prompt does not provide a consistent advantage. For example, models trained with DAPO still perform better under TTRL evaluation than under DAPO

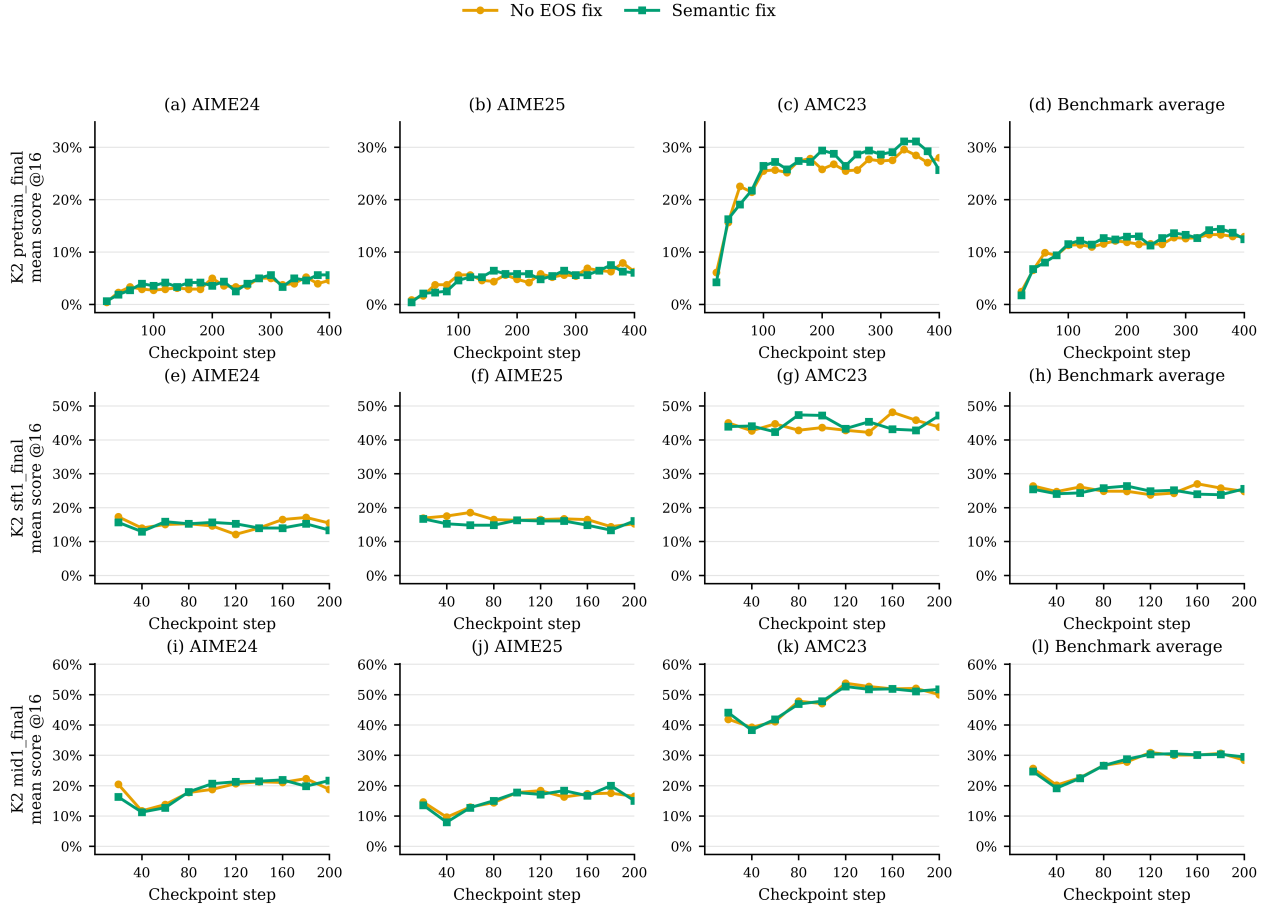


Figure 12 Downstream evaluation with and without semantic EOS correction across K2-Horizon training stages. We report Avg@16 on AIME24, AIME25, and AMC23, together with their unweighted benchmark average, for Pretrain-to-Final, SFT-to-Final, and Midtrain-to-Final distillation. The Pretrain-to-Final runs are evaluated through 400 training steps, while the SFT-to-Final and Midtrain-to-Final runs are evaluated through 200 steps. Across initialization stages and benchmarks, semantic EOS correction yields performance similar to vanilla OPD, with no consistent degradation in downstream accuracy.

evaluation. Since the two evaluation formats differ in both prompting and grading, we interpret this as an effect of the overall evaluation protocol rather than of the prompt alone.

Obs 2: The training prompt induces a persistent response length bias. Models trained with the DAPO template generally produce shorter responses than those trained with the TTRL or raw question templates, and this tendency persists under both evaluation formats. This suggests that the training prompt induces a persistent change in generation behavior that transfers across evaluation prompts.

Obs 3: Response length and measured performance are partially decoupled. The shorter responses induced by DAPO training do not correspond to a consistent reduction in performance. Under TTRL evaluation, models trained with DAPO achieve performance comparable to the other training settings while producing substantially shorter responses. Thus, changes in response length and measured task performance need not move together.

Overall, these results suggest that the training prompt can persistently shape generation behavior, while the evaluation format can substantially affect measured performance. A more systematic characterization of this interaction remains an important direction for ongoing work.

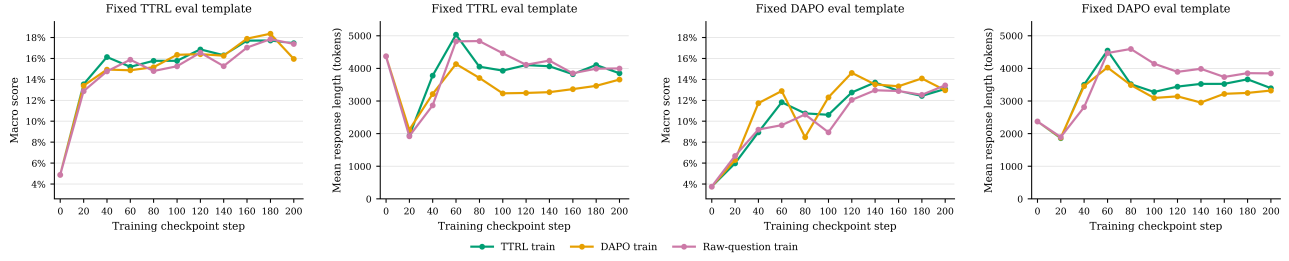


Figure 13 Effect of Training and Evaluation Prompt Templates on Model Behavior. Influence of training and evaluation prompt templates on downstream performance and response length. Models are trained using the TTRL, DAPO, or raw-question template and evaluated with either the TTRL template (left) or the DAPO template (right). The semantic-EOS-class fix is applied in all experiments.

F Rollout Examples from Figure 1b

This section provides the rollouts corresponding to the three examples in Figure 1b. Since each response reaches the full generation budget, we show the reasoning up to the first correct answer together with representative parts of the subsequent repetitive continuation, omitting repeated middle segments for readability.

F.1 Example 1: Answer Repetition

This example is AMC23 problem 10. The correct answer is first produced after 1,094 tokens, followed by 7,098 redundant tokens, accounting for 86.6% of the response.

Prompt

Rows 1, 2, 3, 4, and 5 of a triangular array of integers are shown below.

```
1
1 1
1 3 1
1 5 5 1
1 7 11 7 1
```

Each row after the first row is formed by placing a 1 at each end of the row, and each interior entry is 1 greater than the sum of the two numbers diagonally above it in the previous row. What is the units digits of the sum of the 2023 numbers in the 2023rd row? Please reason step by step, and put your final answer within `\boxed{}`.

Reasoning up to the first correct answer

[...]

```
-> units digit = 8
- $2^8 = 256$ -> units digit = 6
```

We can see that the **units digit of powers of 2** cycles every **4** numbers:

```
- 2, 4, 8, 6, 2, 4, 8, 6, ...
```

So the **pattern repeats every 4 powers**.

Let's find the **position of 2024 in this cycle**:

```
- 2024 / 4 = 506 with a remainder of 0
```

So, the units digit of 2^{2024} is the same as the units digit of 2^4 , which is **6**

Then, the **units digit of $2^{2024} - 1$** is $6 - 1 = 5$

Final Answer:

```
\boxed{5}
```

Repetitive continuation

```
---
### Final Answer:
\boxed{5}
The **units digit of the sum of the numbers in the 2023rd row** is **5**.
---
### Final Answer:
\boxed{5}
---
### Conclusion:
- The **units digit of the sum of the numbers in the 2023rd row** is **5**.
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
[ the '### Final Answer' block above repeats 704 times in total ]
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
---
### Final Answer:
\boxed{5}
---
### Final
[ truncated at the 8192-token budget; no EOS token was emitted ]
```

F.2 Example 2: Self Correction Loop

This example is AIME24 problem 7. The correct answer is first produced after 759 tokens, followed by 7,433 redundant tokens, accounting for 90.7% of the response.

Prompt

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x) = \log_y(x^{4y}) = 10$. Find xy . Please reason step by step, and put your final answer within $\boxed{}$.

Reasoning up to the first correct answer

```
[...]
ifferent bases:

We know that:


$$\log_x(y) = \frac{\log_y(y)}{\log_y(x)} = \frac{1}{\log_y(x)} = \frac{1}{b}$$


So:


$$a = \frac{1}{b} \quad \Rightarrow \quad \frac{10}{x} = \frac{1}{\frac{5}{2y}} = \frac{2y}{5}$$


So:


$$\frac{10}{x} = \frac{2y}{5} \quad \Rightarrow \quad 10 = \frac{2y}{5} \cdot x \quad \Rightarrow \quad 10 = \frac{2xy}{5}$$


Multiply both sides by 5:


$$50 = 2xy \quad \Rightarrow \quad xy = \frac{50}{2} = 25$$


---

### Final Answer:


$$\boxed{25}$$

```

Repetitive continuation

```

$$xy = \boxed{25}$$


---

### Conclusion:

We found that:


$$xy = \boxed{25}$$


**Answer:**  $\boxed{25}$ 

---

### Verification:

We can verify the solution:

Let's assume  $xy = 25$ , and solve for  $x$  and  $y$ .

From  $xy = 25$ , we can choose  $x = 5$  and  $y = 5$ , so  $xy = 25$ .

Now, check the two logarithmic equations:

1.  $\log_x(y^x) = \log_5(5^5) = \log_5(3125) = 5$  (not 10, so this is incorrect).

We need to choose values such t
```

```
[ the 'Let's try' block above repeats 371 times in total ]

et's try $x = 2$, $y = 12.5$.

Let's try $x = 2$, $y = 12.5$.

Let's try $x = 2$, $y = 12.5$.

Let's try $x = 2$, $y = 1

[ truncated at the 8192-token budget; no EOS token was emitted ]
```

F.3 Example 3: Token Repetition

This example is AMC23 problem 0. The correct answer is first produced after 462 tokens, followed by 7,730 redundant tokens, accounting for 94.4% of the response. Two non ASCII symbols in the original rollout are rendered as `[+]` and `[-]` below for typesetting compatibility.

Prompt

Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet? Please reason step by step, and put your final answer within `\boxed{}`.

Reasoning up to the first correct answer

```
[...]
ances they travel** should equal the **total distance between them**, which is 45 miles.

So:

$$
18t + 12t = 45
$$

---

### Step 3: Solve for  $t$ 

$$
30t = 45
$$

$$
t = \frac{45}{30} = 1.5 \text{ hours}
$$

So, they meet after 1.5 hours.

---

### Step 4: Find the distance from City  $A$ 

Alicia is biking towards  $B$  at 18 mph, so in 1.5 hours, she will have traveled:

$$
\text{Distance from } A = 18 \cdot 1.5 = 27 \text{ miles}
$$

---

### Final Answer:

$$
\boxed{27}

```

Repetitive continuation

```
$$

They will be 27 miles from City  $A$  when they meet. [+]
```

$$\boxed{27}$$

[the check-mark token above repeats 3836 times in total]

```
[ truncated at the 8192-token budget; no EOS token was emitted ]
```